

QUESTION1) This problem requires the mathematical modeling of a **Quadrotor yaw-axis** dynamics while it stays in the air.

SOLUTION1) When creating the **transfer function** of a system, the time domain response characteristics, for example the **time constant** are directly changed to the **positions of poles and zeros in the s-plane**.

In this case, the duration contribution information given in the question shows the time constants of the **zeros of the system** and the **numerator of the transfer function** in detail with this information.

Before starting the solution, the principles used are stated:

The numerator **N(s)** made from the **system zeros**,shapes **transient response** and **overshoot/undershoot**.

The relation between the time constant (τ) and the root:

$$|z| = \frac{1}{\tau}$$

(1.1)

Justify: A **large time constant** must show that the mode of the system I study is **very slow and close to the origin** in the s-plane. the opposite, a **small time constant**, represents for me a **fast dynamic and a root far from the origin**.

Minimum phase: **Left-Half-Plane** zeros and poles stabilize.

Non-minimum phase: **Right-Half-Plane** zero exists system.

A **Right-Half-Plane** zero does not affect **stability** but causes brief **undershoot**. E.g, turn right command makes the quadrotor move left first.

This time is the time constant of that zero.

This time is shown as (τ_{nm}) .

$$\tau_{nm} = 30 \text{ second}$$

(1.2)

To determine the location of the **non-minimum phase zero z_{nm}**, the previously mentioned **time constant relationship** and obtained the **magnitude of the zero** from this expression.

$$|z_{nm}| = \frac{1}{\tau_{nm}} = \frac{1}{30} \approx 0.0333$$

(1.3)

Since the system is **not in minimum phase**, the zeros must lie in the **right half-plane**. Therefore, I take the calculated zero as positive, because all roots located in the Right Half-Plane have **positive real parts**.

$$z_1 = +0.0333$$

(1.4)

If a root is at the point z₁, then its factor in the transfer function will be (s−z₁).

The first term of the numerator is:

$$Term \ 1 \rightarrow N_1(s) = (s - 0.0333)$$

(1.4)

It is said in the question that the other **minimum-phase zero** gives **27.7 times less time** than the first zero. Less time means a **faster dynamic** and a **smaller time constant**.

Time constant calculation:

$$\tau_m = \frac{\tau_{nm}}{27.7} = \frac{30}{27.7} \approx 1.083 \text{ second}$$

(1.5)

Calculating the position:

$$|z_m| = \frac{1}{\tau_m} = \frac{1}{1.083} \approx 0.9233$$

(1.6)

Since the zero is **minimum phase** it is located in the **Left-Half-Plane** of the s-plane because zeros of a minimum-phase system must be there.

The real part of the root is **always negative**.

$$z_2 = -0.9233$$

(1.7)

The root is equal to -0.9233 so,the multiplier is also equal to (s-(-0.9233).

$$Term \ 2 \rightarrow N_2(s) = (s + 0.9233)$$

(1.8)

The total numerator is equal to N(s) so,the product of N₁ and N₂ is equal to N(s).

$$N(s) = N_1(s) \cdot N_2(s)$$

(1.9)

$$N(s) = (s - 0.0333)(s + 0.9233)$$

(1.10)

When I expand this expression:

$$N(s) = s^2 + (0.9233 - 0.0333)s - (0.0333 \times 0.9233)$$

(1.11)

Consequently,the numerator equal to:

$$N(s) = s^2 + 0.89s - 0.0307$$

(1.12)

QUESTION2) Analyze the problem,that the **characteristic equation** of the quadrotor's yaw Dynamics denominator of the system is requested to be modeled.

SOLUTION2) In this context the requested **denominator** contains the poles determine of the **system stability** and **transient response**.Periodic trajectory,settling time and **divergent behavior** allow calculation of the poles real and imaginary parts in the s-plane.

Two different dynamic behavior groups are defined in the system. The first one is **convergent** and the second one is **divergent**.

Analysis of Convergent and Oscillatory Poles:

The system has a **damped pair of complex poles**, as it oscillates with a certain period and **reaches equilibrium** within a defined time.

Determination of damped frequency (ω_d):

1) 31.4 second periodic trajectory.

2) **Justify:**In any system, the **oscillation period (T)** in the **step response** is **inversely related** to the **damped natural frequency (ω_d)**.

Calculation:

$$\omega_d = \frac{2\pi}{T} \quad T = 31.4 \text{ s}$$

(2.1)

$$\omega_d = \frac{2 \times 3.14159}{31.4} \approx 0.2 \text{ rad / s}$$

(2.2)

The **imaginary part** causes the system to **oscillate**. If the imaginary part were 0 the system would **settle directly without** any oscillation (**overdamped**).

Determination the real part(σ):

1) Settles in 43 second.

2) The settling time is usually calculated with the **4 time constant (4τ)** criterion because it happens when the **error goes below 2%** and the **system reaches steady-state**. e^(-t/τ) show the **transient decay**.With 4 time constant, the system error becomes 1.8% and this value is the first integer inside the 2% safety limit.

$$\text{for } t = 4\tau \quad e^{-4} = 0.018$$

(2.3)

This formula is as follows:

$$t_s \approx \frac{4}{\sigma} \text{ (where } \sigma = \zeta \omega_n \text{)}$$

(2.4)

Calculation:

$$\sigma = \frac{4}{t_s} \quad \sigma = \frac{4}{43} \approx 0.0930$$

(2.5)

Since the system is **stable** and exhibits periodic behavior the poles are known to **have real and imaginary parts**.For stability,poles must lie in the **Left-Half-Plane**,so a negative sign is placed before 0.0930

$$\text{Real part} = -0.0930$$

(2.6)

Combining results:

The system oscillates and is damped because it has **real and imaginary parts** real causes **damping**,imaginary causes **oscillation**. Complex poles in physical systems always appear as **conjugate pairs**, with +jand -j.

Therefore, the poles are written in this form:

$$s = (\text{Real part}) \pm j(\text{Imaginary part})$$

(2.7)

Since the system is **convergent**, the **poles are in the left half plane**. This means that the **real part** is **negative**.

$$s_{1,2} = -0.0930 \pm j0.2$$

(2.8)

Polynomial of the Convergent Part:The polynomial created by these **conjugate roots**

If the roots of a system are s_{1,2} = −σ + jω_d, the characteristic equation has the following form:

$$(s - s_1) \cdot (s - s_2) = (s + \sigma)^2 + \omega_d^2$$

(2.9)

This is verified by substituting the numbers: s_{1,2} = −0.093 ± j0.2.

$$\text{Polynomial: } P(s) = (s - s_1) \cdot (s - s_2)$$

(2.10)

$$s_1 = -0.093 + j0.2 \quad s_2 = -0.093 - j0.2$$

(2.11)

When substitute the s₁ and s₂ into the equation:

$$P(s) = (s - (-0.093 + j0.2)) \cdot (s - (-0.093 - j0.2))$$

(2.12)

The parentheses are opened applied the negative signs:

$$P(s) = (s + 0.093 - j0.2) \cdot (s + 0.093 + j0.2)$$

(2.13)

Since s + 0.093 appears on both sides, I named it A. I named the j0.2 part B.

The equation now takes the following form:

$$P(s) = (A - B) \cdot (A + B)$$

(2.14)

By using the difference of squares formula ((x − y)(x + y) = x² − y²):

$$P(s) = A^2 - B^2$$

(2.15)

Then, replaced A and B with their original expression:

The part corresponding to A²:

$$A^2 = (s + 0.093)^2$$

(2.16)

The perfect square expansion rule ((x + y)² = x² + 2xy + y²):

$$A^2 = s^2 + 2(0.093)s + (0.093)^2 \quad A^2 = s^2 + 0.186s + 0.008649$$

(2.17)

The part corresponding to B²:

$$B^2 = (j0.2)^2 \quad B^2 = j^2 \cdot (0.2)^2$$

(2.18)

Because j² = −1 :

$$B^2 = (-1) \cdot 0.04 = -0.04$$

(2.19)

Conclusion:

$$P(s) = A^2 - B^2 \quad P(s) = (s^2 + 0.186s + 0.008649) - (-0.04)$$

(2.20)

$$P(s) = s^2 + 0.186s + 0.008649 + 0.04$$

(2.21)

By adding the constants:

$$0.008649 + 0.04 \approx 0.0486$$

(2.22)

$$P(s) = D_{nm}(s) = s^2 + 0.186s + 0.0486$$

(2.23)

Divergent Pole Analysis:

It is stated that the **system is not only stable but also unstable**.Divergent character which is located **78.5 times farther than the convergent pole position**.

Justify: By saying divergent, understood that the pole is in the **Right-Half-Plane** of the s-plane, meaning it is a positive real root.

The expression "**convergent pole position**" is equal to the distance of the convergent pole on the real axis, which is **σ**.

Calculation:

1) The system initially settles slowly, with a **small distance** of 0.093.

2) The system diverges quickly and strongly, with a **long distance** of 7.3.

The **divergent pole** is found by multiplying the **smaller distance** by **78.5** distance from origin on real axis.

$$p_{div} = 78.5 \times \sigma = 78.5 \times 0.0930 \approx 7.3$$

(2.24)

Pole position: Since it is divergent, took the sign as positive.

$$p_s = +7.3$$

(2.25)

Divergent Part Polynomial: Since the root is +7.3 took the factor as (s−7.3).

$$D_{div}(s) = (s - 7.3)$$

(2.26)

Conclusion:

The system's characteristic equation (denominator) is obtained by **multiplying the convergent and divergent polynomials**:

$$D(s) = D_{conv}(s) \cdot D_{div}(s)$$

(2.27)

$$D_{conv}(s) = (s^2 + 0.186s + 0.0486)$$

(2.28)

$$D_{div}(s) = (s - 7.3)$$

(2.29)

$$D(s) = (s^2 + 0.186s + 0.0486)(s - 7.3)$$

(2.30)

General Transfer Function (Final Equation):

When the **obtained numerator** and the **denominator** the mathematical model of the system appears as follows:

$$G(s) = \frac{N(s)}{D(s)} = \frac{s^2 + 0.89s - 0.0307}{(s^2 + 0.186s + 0.0486) \cdot (s - 7.3)}$$

(2.31)

$$G(s) = \frac{N(s)}{D(s)} = \frac{s^2 + 0.89s - 0.0307}{s^3 - 7.114s^2 - 1.309s - 0.355}$$

(2.32)

The transfer function G(s) has two parts , the **unstable pole** and the **stable oscillatory pole pair**.ζ, ω_n and ω_d are found from the second.

This part of the denominator is:

$$s^2 + 0.186s + 0.0486$$

(2.33)

Standard second-order equation form:

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

(2.34)

Natural Frequency (ω_n):In the standard form, the constant term is ω_n².

$$\omega_n^2 = 0.0486$$

(2.35)

$$\omega_n = \sqrt{0.0486} \approx 0.2205 \text{ rad / s}$$

(2.36)

Damping Ratio (ζ):In the standard form, the **coefficient of s** is 2ζω_n. It is also called "zeta".

$$2\zeta\omega_n = 0.186$$

(2.37)

$$\zeta = \frac{0.186}{2 \times 0.2205} \approx 0.422$$

(2.38)

Damped Natural Frequency (ω_d):The damped frequency is the frequency at which the system actually oscillates.

Formula:

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

(2.39)

$$\omega_d = 0.2205 \times \sqrt{1 - 0.422^2}$$

(2.40)

$$\omega_d \approx 0.20 \text{ rad / s}$$

(2.41)

Damping Type: Looked at **zeta** to understand the **system behavior** because zeta is between 0 and 1 the system is **underdamped**.

$$0 < \zeta < 1: \text{ Underdamped (less damping)}$$

(2.42)

With this result, the system oscillates before reaching equilibrium.

Maximum Overshoot:

Formula:

$$\%OS = e^{\frac{-\zeta\omega_n}{\sqrt{1-\zeta^2}}} \times 100$$

(2.43)

OS shows how many percent the output goes higher than the final value.

When put the values in:

$$\%OS = e^{\frac{-0.422 \times 0.2}{\sqrt{1 - 0.422^2}}} \times 100 \approx \%23.18$$

(2.44)

Y(s) Expression (Step Response):When a unit step input (R(s)=1/s) is applied the **output function Y(s)** is obtained.

When the equation is put into it,

$$Y(s) = \frac{(s - 0.0333)(s + 0.9233)}{s(s^2 + 0.186s + 0.0486)(s - 7.3)}$$

(2.45)

$$Y(s) = \frac{s^2 + 0.89s - 0.03075}{s(s^3 - 7.114s^2 - 1.309s - 0.355)}$$

(2.46)

The ζ and %OS values show only **oscillation** but the **unstable pole** at s=7.3 causes the system to **diverge to infinity** with the **overshoot** appearing as a **small wave**.

QUESTION3) In this situation, the **quadrotor yaw dynamics** modeled in Questions1 and 2 into a **P-controller** by using a **proportional gain** (pure gain).

SOLUTION3) In this case, while doing the analysis, a **brute-force stability check** was used. In this specific scenario, there is a **very critical 'root trap'** that prevents the system from becoming **stable**. According to the **root locus rule**, the **closed-loop poles of a system move from the open-loop poles toward the open-loop zeros** as the proportional gain **K_p** increases.To analyze the stability of the system the **denominator**

of the **closed-loop transfer function**, which is the **characteristic equation**. For mathematical stability, **all coefficients** of the characteristic equation **must have the same sign** and the system must also satisfy the **Routh–Hurwitz criteria**.

Forming the characteristic equation

The closed-loop system's characteristic equation:

$$1 + K_p G(s) = 0$$

(3.1)

Here, because $G(s) = \frac{N(s)}{D(s)}$ it should be written by replacing it into the equation:

$$D(s) + K_p N(s) = 0$$

(3.2)

Denominator and numerator polynomials:

$$D(s) (\text{denominator}) \rightarrow s^3 - 7.114s^2 - 1.309 - 0.355$$

$$N(s) (\text{numerator}) \rightarrow s^2 + 0.89s - 0.0307$$

(3.3)

When the equations substituted:

$$(s^3 - 7.114s^2 - 1.309s - 0.355) + K_p (s^2 + 0.89s - 0.0307) = 0$$

(3.4)

When the polynomials are grouped:

$$\text{The } s^3 \text{ term: It only comes from } D(s) \rightarrow 1 \cdot s^3$$

$$\text{The } s^2 \text{ term: } -7.114 + K_p \rightarrow (K_p - 7.114)s^2$$

(3.5)

$$\text{The } s^1 \text{ term: } -1.309 + 0.89K_p \rightarrow (0.89K_p - 1.309)s$$

(3.6)

$$\text{The } s^0 (\text{constant}) \text{ term: } -0.355 - 0.0307K_p \rightarrow -(0.355 + 0.307K_p)$$

(3.6)

As a result, the characteristic equation is:

$$s^3 + (K_p - 7.114)s^2 + (0.89K_p - 1.309)s - (0.355 + 0.03075K_p) = 0$$

(3.7)

Necessary Condition Check

According to the **Routh-Hurwitz** rule all **characteristic equation coefficients** must be **non-zero** and have the **same sign**, since the highest-degree coefficient is **positive** all other must also positive.

When the coefficients were examined,

Analysis of the s² coefficient is:

$$(K_p - 7.114) > 0 \text{ should be true} \quad K_p > 7.114$$

(3.8)

Analysis of the s⁰ (constant) coefficient (Critical Point):

$$Term \rightarrow -(0.355 + 0.03075K_p)$$

(3.9)

This term must be positive,

$$-(0.355 + 0.03075K_p) > 0$$

(3.10)

When the equation is solved,

$$0.355 + 0.03075K_p < 0 \quad 0.03075K_p < -0.355 \quad K_p < -11.54$$

(3.11)

Under these circumstances,mathematically impossible situation because:

For the system to be stable, the **s² term requires K_p > 7.114** (a positive large gain) but the **s⁰ term requires K_p < −11.54** (a negative gain).

It is **not possible** for a number to be greater than 7 and less than -11 at the same time.Such a situation cannot

QUESTION4) The purpose of this question is to design a PI controller to stabilize the system.

SOLUTION4) In this situation, PI controllers are used often in industry because the integral action helps the system follow the reference signal with zero steady-state error but in this case, the integral action adds phase lag to the system and this usually makes the stability margins worse.

The system modeled in Question 1 and Question 2 has a very strong and aggressive instability.

Adding a PI controller increases the order of the system and also makes the relationship between the coefficients more complex.

The denominator of the closed-loop transfer function (the characteristic equation):

$$1 + C_{PI}(s)G_{plant}(s) = 0$$

(4.1)

The transfer function of the PI controller:

$$C_{PI}(s) = K_p + \frac{K_i}{s} = \frac{K_p s + K_i}{s}$$

(4.2)

The transfer function of the Plant:

$$G_{plant}(s) = \frac{N(s)}{D(s)} = \frac{s^2 + 0.89s - 0.0307}{s^3 - 7.114s^2 - 1.309s - 0.355}$$

(4.3)

When it it put into the equation:

$$1 + \left(\frac{K_p s + K_i}{s}\right) \left(\frac{N(s)}{D(s)}\right) = 0$$

(4.4)

When the denominators are made the same and simplified:

$$s \cdot D(s) + (K_p s + K_i) \cdot N(s) = 0$$

(4.5)

When the polynomials are grouped:

The system became fourth-order because its order increased by one:

$$1^{st} \text{ Term}(s \cdot D(s)) \rightarrow s(s^3 - 7.114s^2 - 1.309s - 0.355) = s^4 - 7.114s^3 - 1.309s^2 - 0.355s$$

(4.6)

$$2^{nd} \text{ Term} \rightarrow ((K_p s + K_i) \cdot N(s)) \text{ When I expand this product as } (K_p s + K_i)(s^2 + 0.89s - 0.0307s)$$

(4.7)

$$\text{The } s^3 \text{ Term: } K_p \cdot s^2 \cdot s = K_p s^3$$

(4.8)

$$\text{The } s^2 \text{ Term: } K_p(0.89s^2) + K_i(s^2) = (0.89K_p + K_i)s^2$$

(4.9)

$$\text{The } s^1 \text{ Term: } K(-0.0307s) + K_i(0.89s) = (0.89K_i - 0.0307K_p)s$$

(4.10)

$$\text{The } s^0 \text{ (constant) Term: } K(-0.0307) = -0.0307K_i$$

(4.11)

Adding the second term creates a fourth-order equation:

$$s^4 + (K_p - 7.114)s^3 + (0.89K_p + K_i - 1.309)s^2 + (0.89K_i - 0.0307K_p - 0.355)s - 0.0307K_i = 0$$

(4.12)

For stability all characteristic equation coefficients must be non-zero and positive.

When the coefficients are examined,

$$\text{The coefficient of } s^3(K_p - 7.114) > 0 \rightarrow K_p > 7.114 \quad \text{Constant Term } s^0 \text{ (Critical Issue): } -0.0307K_i$$

$$\text{This means we need a positive and large } K_p \quad \text{For this term to be positive: } -0.0307K_i > 0$$

(4.13)

When the inequality is divided by -0.0307s:

$$K_i < 0$$

(4.14)

In PI design K_p and K_i must be positive but making K_i negative to stabilize the system break control,causes positive feedback and invalidates intermediate coefficient sign.

Justify:The new pole shifts the open-loop root locus right,increasing instability.Due to RHP pole-zero the s^0 term is negative so a PI controller with positive K_i fails to stabilize the non-minimum phase system everything remains unstable.

```
s = tf('s'); % Laplace variable
n = [1 0.89 -0.0307]; % numerator
d = [1 -7.114 -1.309 -0.355]; % denominator
G = tf(n, d);
sp_vec = zeros(1,100);
Kp_vec = -0.001:1.000;
Ki_vec = -0.001:1.000;
stab = zeros(length(Kp_vec), length(Ki_vec)); % stability map
for i = 1:length(Kp_vec)
    for j = 1:length(Ki_vec)
        Kp = Kp_vec(i);
        Ki = Ki_vec(j);
        % Transfer function
        % C(s) = (Kp*s + Ki)/s
        % Closed-loop transfer function
        % T(s) = (Kp*s + Ki) * G(s) / (1 + (Kp*s + Ki) * G(s))
        % Characteristic equation
        % 1 + (Kp*s + Ki) * G(s) = 0
        % Simplify the equation
        % s^4 + (Kp - 7.114)s^3 + (0.89Kp + Ki - 1.309)s^2 + (0.89Ki - 0.0307Kp - 0.355)s - 0.0307Ki = 0
        % Find roots
        % roots([Kp - 7.114, 0.89Kp + Ki - 1.309, 0.89Ki - 0.0307Kp - 0.355, -0.0307Ki])
        % Check stability
        % all(real(poles) < 0)
        stab(i,j) = 1;
    end
end
end
% Plot map
figure;
imagesc(Ki_vec, Kp_vec, stab);
title('PI Controller Stability Map');
xlabel('Ki');
ylabel('Kp');
```

Figure 3 PI-Controller Stability Map

After the simulation, the blue area shows that the PI controller cannot make the system stable when K_p and K_i have different positive or negative values.Due to Mathematical requirements, the negative term needed to correct the constant term destroys stability therefore,the system is not stabilizable using a PI controller.

```
% Plant parameters
n_plant = [1 0.89 -0.0307]; % numerator
d_plant = [1 -7.114 -1.309 -0.355]; % denominator
G_plant = tf(n_plant, d_plant);
% Controller parameters
Kp_list = [0.5 1 2];
Ki_list = [0.2 0.5 1 2];
t = 0:0.01:5;
% Step response
figure;
hold on;
grid on;
legend_text = [];
for i = 1:length(Kp_list)
    for j = 1:length(Ki_list)
        Kp = Kp_list(i);
        Ki = Ki_list(j);
        % Closed-loop transfer function
        % C(s) = (Kp*s + Ki)/s
        % System transfer function
        % Sys_CL = Feedback(C_pl * G_plant, 1)
        % Step response
        y = step(Sys_CL, t);
        plot(t, y, 'linewidth', 3);
        legend_text(end+1) = [Kp', num2str(Kp), ', ', Ki', num2str(Ki)];
    end
end
title('Step Response for Different Kp and Ki');
xlabel('Time (s)');
ylabel('Output');
```

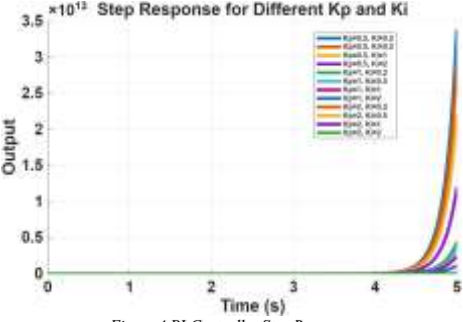


Figure 4 PI-Controller Step Response

QUESTIONS) The purpose of this question is to design the system with a PD controller.

SOLUTIONS) In this situation, to handle the strong natural instability, looking at the error (P) or past error (I) is not effective. Therefore, the derivative action, which predicts the system's tendency and provides virtual damping, should be used. Designing with a PD controller adds things to the system like zero addition, locus reshaping and damping. PD design, K_p and K_d ensure stability RHP zero causes initial undershoot closed-loop roots must be negative.

The first goal is to adjust the coefficients of the unstable third-order characteristic equation with the PD controller parameters and move the roots into the Left-Half-Plane.

Plant(G(s):

$$G(s) = \frac{N(s)}{D(s)} = \frac{s^2 + 0.89s - 0.03075}{s^3 - 7.114s^2 - 1.309s - 0.355}$$

(5.1)

PD Controller(C(s)):

$$C(s) = K_p + K_d s$$

(5.2)

The transfer function of the closed-loop determines stability.

$$1 + C(s)G(s) = 0$$

(5.3)

When it is substituted into the equation:

$$1 + (K_p + K_d s) \left(\frac{s^2 + 0.89s - 0.03075}{s^3 - 7.114s^2 - 1.309s - 0.355} \right) = 0$$

(5.4)

When the denominators are made equal and simplified:

$$\frac{(s^3 - 7.114s^2 - 1.309s - 0.355) + (K_d s + K_p)(s^2 + 0.89s - 0.03075)}{\text{Plant Denominator } D(s)} = 0$$

(5.5)

The second part of the equation, which is the multiplication of the PD controller and the plant denominator:

$$1 + (K_p + K_d s) \left(\frac{s^2 + 0.89s - 0.03075}{s^3 - 7.114s^2 - 1.309s - 0.355} \right) = 0$$

(5.6)

Multiplication with $K_d s$:

$$K_d s \cdot s^2 \rightarrow K_d s^3 \quad K_d s \cdot 0.89s \rightarrow 0.89K_d s^2 \quad K_d s \cdot (-0.03075) \rightarrow -0.03075K_d s$$

(5.7)

Multiplication with K_p :

$$K_p \cdot s^2 \rightarrow K_p s^2 \quad K_p \cdot 0.89s \rightarrow 0.89K_p s \quad K_p \cdot (-0.03075) \rightarrow -0.03075K_p$$

(5.8)

Sum of the equation:

Adding the plant denominator and the result of the multiplication, and grouping the terms with the same degree:

Equation:

$$D(s) + [PD \times N(s)] = 0$$

(5.9)

s^3 Term:	s^2 Term:
Original: s^3	Original: $-7.114s^3$
From PD: $K_d s^3$	From PD: $0.89K_d s^2 + K_p s^2$
Total: $(1 + K_d)s^3$	Total: $(K_p + 0.89K_d - 7.114)s^2$
s^2 Term: <td>s^1 (Constant) Term:</td>	s^1 (Constant) Term:
Original: $-1.309s$	Original: -0.355
From PD: $-0.03075K_p + 0.89K_d s$	From PD: $-0.03075K_p$
Total: $(0.89K_d - 0.03075K_p - 1.309)s$	Total: $(-0.03075K_p + 0.355)$

(5.10)

The Closed-Loop Characteristic Equation obtained is:

$$(1 + K_d)s^3 + (K_p + 0.89K_d - 7.114)s^2 + (0.89K_d - 0.03075K_p - 1.309)s - (0.03075K_p + 0.355) = 0$$

(5.12)

Under these circumstances, the characteristic equation must satisfy some mathematical requirements to be stable:

Coefficient of s^3 :

$$1 + K_d > 0 \rightarrow K_d > -1 \text{ (Generally, } K_d \text{ is positive.)}$$

(5.13)

Coefficient of s^2 (critical):

$$K_p + 0.89K_d - 7.114 > 0$$

(5.14)

Actually, the coefficient was unstable. With the PD controller, K_p and K_d were added. If K_p and K_d are chosen big enough, the term becomes positive. This is the mathematical effect of damping.

Coefficient of s^0 :

$$-(0.03075K_p + 0.355) > 0$$
$$K_p < -\frac{0.355}{0.03075} \approx -11.54$$

(5.15)

(5.16)

Justify:With the zero added in the Left-Half-Plane and the derivative effect introduced by the PD controller, the root locus trapped between the unstable pole and the zero in the RHP is reshaped and pulled into the stable region.

Conclusion:

With the P controller only K_p is used, so the equation did not improve but with the PD controller, K_d increases the coefficients of the s^2 and s^3 terms and pushes the roots to the LHP.

The K_p and K_d combinations that satisfy all inequalities are in the yellow area on the graph.

```
% Plant parameters
n2 = 1;
d2 = [1 -7.114 -1.309 -0.355];
% PD Controller parameters
Kd_list = [0.001 0.01 0.1 1 10];
Kp_list = [0.001 0.01 0.1 1 10];
% Stability matrix
stab = zeros(length(Kd_list), length(Kp_list));
% Main loop
for i = 1:length(Kd_list)
    for j = 1:length(Kp_list)
        Kd = Kd_list(i);
        Kp = Kp_list(j);
        % Closed-loop transfer function
        % C(s) = Kp + Kd*s
        % System transfer function
        % Sys_CL = Feedback(C * G, 1)
        % Step response
        % roots([Kd, Kp, -7.114, -1.309, -0.355])
        % Check stability
        % all(real(poles) < 0)
        stab(i,j) = 1;
    end
end
% Plot
figure(1);
imagesc(Kp_list, Kd_list, stab);
title('PD Controller Stability Map');
xlabel('Kp');
ylabel('Kd');
```

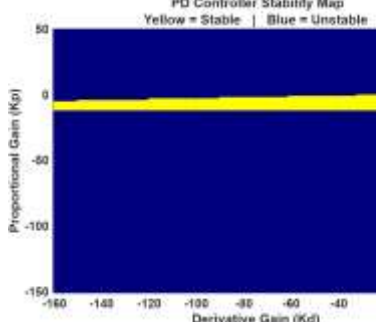


Figure 5 PD-Controller Stability Map

The mathematical limits of the system show that stability only happens when K_p and K_d are negative because of this, the chosen ranges were optimized by removing The diagonal stability area, which is not wide, was then shown clearly.

```
% PD Controller parameters
Kd_list = [0.001 0.01 0.1 1 10];
Kp_list = [0.001 0.01 0.1 1 10];
% Stability matrix
stab = zeros(length(Kd_list), length(Kp_list));
% Main loop
for i = 1:length(Kd_list)
    for j = 1:length(Kp_list)
        Kd = Kd_list(i);
        Kp = Kp_list(j);
        % Closed-loop transfer function
        % C(s) = Kp + Kd*s
        % System transfer function
        % Sys_CL = Feedback(C * G, 1)
        % Step response
        % roots([Kd, Kp, -7.114, -1.309, -0.355])
        % Check stability
        % all(real(poles) < 0)
        stab(i,j) = 1;
    end
end
% Plot
figure(1);
imagesc(Kp_list, Kd_list, stab);
title('PD Controller Stability Map');
xlabel('Kp');
ylabel('Kd');
```

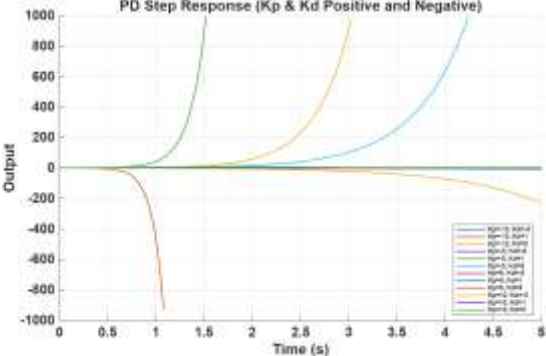


Figure 6 PD-Controller Step Response

QUESTION6) Design PD controller for best system performance.

SOLUTION6) In this situation, to get the best performance with the PD controller, the settling time, overshoot, and divergence of the system are checked because of the RHP zero, even if the system is tuned very well, the output makes an initial overshoot. The best controller only make this overshoot very small or make the settling time shorter.

The goal is to move the closed-loop dominant poles to the best and far point in the left LHP to make the settling time shorter.

- To optimize the PD controller, look at K_p and K_d .
- K_d predicts the future error and adds damping.
- When K_p increases, the system becomes faster, but if K_p is too high, overshoot happens.
- There is a direct relationship between K_d and K_p .

Plant(G(s)):

$$G(s) = \frac{N(s)}{D(s)} = \frac{s^2 + 0.89s - 0.03075}{s^3 - 7.114s^2 - 1.309s - 0.355}$$

(6.1)

PD Controller(C(s)):

$$C(s) = K_p + K_d s$$

(6.2)

When the expression for G(s) and C(s) are put into the equation:

$$1 + (K_p + K_d s) \left(\frac{s^2 + 0.89s - 0.03075}{s^3 - 7.114s^2 - 1.309s - 0.355} \right) = 0$$

(6.3)

After the denominators are made equal the closed-loop characteristic polynomial A(s) is obtained:

$$A(s) = \frac{(s^3 - 7.114s^2 - 1.309s - 0.355) + (K_d s + K_p)(s^2 + 0.89s - 0.03075)}{D(s) \times N(s)} = 0$$

(6.4)

The terms are taken into the s bracket and grouped and the third-order equation that forms the optimization is obtained:

$$\Delta(s, K_p, K_d) = a_3 s^3 + a_2 s^2 + a_1 s + a_0 = 0$$

(6.5)

The coefficients in terms of the Kp and Kd parameters:

$$\begin{aligned} s^3 \text{ Term}(a_3) &= 1 + K_d \\ s^2 \text{ Term}(a_2) &= -7.114 + 0.89K_d + K_p \\ s^1 \text{ Term}(a_1) &= -1.309 - 0.03075K_d + 0.89K_p \\ s^0 \text{ Term}(a_0) &= -0.355 - 0.03075K_p \end{aligned}$$

(6.6)

Under these circumstances, the roots of the polynomial are calculated. Usually, one pair of complex conjugate roots and one real root appear.

$$s_{1,2} = -\sigma \pm j\omega_d$$

(6.7)

The parameters that determine the system's performance:

- Settling Time (t_s), Real part: (σ): It determines how fast the system reaches its steady state.

$$t_s \approx \frac{4}{\sigma_{dominant}}$$

(6.8)

- Overshoot (%OS), Damping Ratio (ζ):

$$\zeta = \frac{\sigma}{\sqrt{\sigma^2 + \omega_d^2}} \quad M_p = \%OS = 100 \cdot e^{-\frac{\zeta \omega_n}{\sqrt{1-\zeta^2}}}$$

(6.9)

```
% PD Controller parameters
Kd_list = [0.001 0.01 0.1 1 10];
Kp_list = [0.001 0.01 0.1 1 10];
% Stability matrix
stab = zeros(length(Kd_list), length(Kp_list));
% Main loop
for i = 1:length(Kd_list)
    for j = 1:length(Kp_list)
        Kd = Kd_list(i);
        Kp = Kp_list(j);
        % Closed-loop transfer function
        % C(s) = Kp + Kd*s
        % System transfer function
        % Sys_CL = Feedback(C * G, 1)
        % Step response
        % roots([Kd, Kp, -7.114, -1.309, -0.355])
        % Check stability
        % all(real(poles) < 0)
        stab(i,j) = 1;
    end
end
% Plot
figure(1);
imagesc(Kp_list, Kd_list, stab);
title('PD Controller Stability Map');
xlabel('Kp');
ylabel('Kd');
```

Figure 7 Best PD-Controller 3D Optimization Landscape

In the simulation, the poles of the equation were calculated using root analysis and the gains were combined with system performance. The real part of dominant pole was maximized to reduce settling time and unstable points were filtered out. The marked point represent the safe K_p and K_d values that provide the fastest response.